SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 35%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Question 1: Doctor Shift Assignment Using Backtracking Search

Problem: Assign doctors D1, D2 and D3 to Morning, Afternoon and Night shifts, with exactly one doctor per shift.

Constraints:

- D1 cannot work the Night shift.
- D2 must work before D3.
- Only one doctor can be assigned to each shift.
- D3 cannot work the Morning shift.
- Every doctor must receive exactly one shift.

Backtracking steps and constraint checks

Step 1: Assign D1. Try Morning. This satisfies the rule that D1 cannot work Night. Keep D1 = Morning.

Step 2: Assign D2. Try Afternoon. The shift is not occupied, so the one-doctor-per-shift constraint is satisfied.

Step 3: Assign D3. Assign Night. D3 is not in Morning, and D2 (Afternoon) is before D3 (Night).

Step 4: Check all constraints. All three shifts are occupied once; all doctors have exactly one shift; D1 is not at Night; D3 is not at Morning; and D2 is before D3. No backtracking is required for this assignment.

Shift	Assigned doctor
Morning	D1
Afternoon	D2
Night	D3

Final valid schedule: Morning – D1; Afternoon – D2; Night – D3.

Question 2: Grid Navigation and Search

Note: The assessment text does not specify the exact locations of the obstacles, Start (S), or Goal (G). For a worked example, the following 5 × 5 grid is assumed. The result is optimal for this example grid.

S	·	X	·	·
·	·	X	·	X
X	·	·	·	·
·	X	X	X	·
·	·	·	·	G

Legend: S = Start, G = Goal, X = obstacle, · = free cell. Movement is allowed only up, down, left, or right. Each move costs 1.

Search method

Breadth-First Search (BFS) explores nodes in increasing order of depth. Because every move has the same cost (1), BFS finds a path with the minimum number of moves. A FIFO queue is used. The Manhattan distance h = |row − goal row| + |column − goal column| can be calculated as a heuristic, but standard BFS does not use the heuristic f e. A heuristic-prioritized search would be Greedy Best-First Search or A*; this soluti equested BFS method.

106%

Step-by-step BFS expansion

1. Start at (1,1). Queue: [(1,1)]. Mark it visited.
2. Expand (1,1): add accessible neighbours (1,2) and (2,1). Queue: [(1,2), (2,1)].
3. Expand (1,2): (1,3) is blocked; add (2,2). Queue: [(2,1), (2,2)].
4. Expand (2,1): (3,1) is blocked; (2,2) is already visited. Queue: [(2,2)].
5. Expand (2,2): add (3,2). Queue: [(3,2)].
6. Expand (3,2): add (3,3). Queue: [(3,3)].
7. Expand (3,3): add (3,4). Queue: [(3,4)].
8. Expand (3,4): add (2,4) and (3,5). Queue: [(2,4), (3,5)].
9. Expand (2,4): (1,4) is added; (2,5) is blocked. Queue: [(3,5), (1,4)].
10. Expand (3,5): add (4,5). Queue: [(1,4), (4,5)].
11. Expand (1,4): add (1,5). Queue: [(4,5), (1,5)].

12. Expand (4,5): add (5,5), the goal. Queue: [(1,5), (5,5)]. The goal is reached when dequeued; its predecessor chain gives the path.

One shortest path is (1,1) → (1,2) → (2,2) → (3,2) → (3,3) → (3,4) → (3,5) → (4,5) → (5,5). Total cost = 8 moves × 1 = 8.

Important: Since the original question does not show a defined grid or obstacle coordinates, use the grid supplied by your instructor if it differs. The path and cost will then need to be recalculated.

Question 3: Autonomous Rescue Robot in a Partially Observable Environment

(a) Analysis of constraints

- Movement: The robot can move only up, down, left, or right, so diagonal movement is not allowed.
- Obstacles: Blocked cells must never be entered. The robot must choose a route around them.
- Limited sensing: The robot can observe only adjacent cells, so it cannot know the full map initially.
- Movement cost: Each normal move costs 1.
- Risky zones: Entering a risky cell adds 2 to the normal cost, making that move cost 3.
- Dynamic knowledge: After sensing new cells, the robot updates its internal map and replans if necessary.
- Safety and cost: The robot should avoid blocked cells and prefer a safe route with the lowest accumulated cost.

(b) Proposed solution: Online Search Agent with Uniform Cost Search

13. Initialize the robot at S with total cost 0 and mark its immediate surroundings as unknown.
14. Sense the adjacent cells and classify each as free, blocked, or risky.
15. Update the internal map using the new observations.
16. Use Uniform Cost Search (UCS) on the known map. Keep candidate positions in a priority queue ordered by accumulated path cost g(n).
17. For each legal neighbour, calculate new cost = current cost + 1 for a normal cell, or current cost + 3 when entering a risky cell.
18. Do not add blocked cells. If a cheaper route to a position is found, update its cost and predecessor.
19. Move one step along the lowest-cost known route, sense again, and update the map.
20. If a newly observed obstacle blocks the planned route, run the search again using the updated map.

21. Stop when the robot reaches G. Reconstruct the route using predecessor information and report the total cost.

UCS is suitable because it considers accumulated cost rather than merely the number of moves. With non-negative movement costs, it finds the least-cost route on the map available to the search. In a partially observable setting, the robot must repeat sensing and planning as it learns more about the environment.

(c) AI concepts demonstrated

Intelligent agent: The robot senses its surroundings, updates its internal representation, chooses an action, and acts to reach the survivor.

Rationality: A rational robot chooses the action expected to minimize travel cost while avoiding known obstacles and unnecessary risk, based on the information currently available.

Environment type: The environment is partially observable, sequential, dynamic from the robot' s point of view as new information is discovered, and includes discrete grid states and actions. It is also cost-sensitive because risky zones have higher costs.

(d) Justification of the chosen approach

Uniform Cost Search is appropriate because each action has a cost and risky zones increase the cost of a route. Unlike BFS, which minimizes the number of steps when all steps have equal cost, UCS minimizes total accumulated cost. Combining UCS with an online agent allows the robot to sense nearby cells, update its knowledge, and replan when it discovers obstacles. The robot should not treat unknown cells as confirmed safe; it should sense and update its plan as it proceeds.

Conclusion

Backtracking finds a valid doctor-to-shift assignment by checking constraints. BFS finds a shortest path when every move has equal cost, while UCS is better for the rescue-robot scenario because normal and risky moves have different costs and therobot learns the map incrementally.

Student declaration: I certify that this submission is my own work and that I have followed the academic integrity guidelines.

Signature: _____